# SIMATS ENGINEERING
## ASSESSMENT TOOL 2

**COURSE NAME: COMPUTER NETWORKS**
**COURSE CODE: CSA07**

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**COURSE OUTCOME COVERED**

***CO2:*** Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

*Assessment Tool Weightage: 30%*

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**QUESTIONS: 5** **Total Marks:50**

Annexure A

# Scenario-Based

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***Code of Conduct:***

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

Annexure A

# Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:
School of Computing → /25
School of Electronics → /26
School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:
Software Development → **/24**
Quality Assurance → **/25**
Human Resources → **/26**
Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

**2001:0db8:abcd:0000:0000:0000:0000:0000/64**

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

**192.168.2.45**

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

## RUBRICS FOR CALCULATION

| Scenario Based Assessment | | | | | |
|---|---|---|---|---|---|
| **Criteria** | **Weightage (%)** | **Level 4 (Exemplary)** | **Level 3 (Proficient)** | **Level 2 (Developing)** | **Level 1 (Beginning)** |
| Understanding of Scenario | 20% | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps | Partial understanding; misses key aspects | Misinterprets or fails to understand the scenario |
| Identification of Key Issues | 20% | Accurately identifies all relevant issues and factors involved | Identifies most key issues with minor omissions | Identifies limited issues; some irrelevant points | Unable to identify key issues |
| Application of Concepts | 25% | Applies appropriate concepts effectively to address the scenario | Applies concepts with minor errors | Limited or partially correct application | Unable to apply relevant concepts |
| Decision Making | 20% | Makes well-reasoned, logical decisions with strong rationale | Makes reasonable decisions with some justification | Makes weak decisions with limited justification | No clear decisions made or rationale provided |
| Clarity of Response | 15% | Response is clear, structured, and logically presented | Generally clear with minor issues | Somewhat unclear or poorly structured | Disorganized and difficult to understand |

## SCENARIO 1: SMART CAMPUS – IPv4 SUBNET ALLOCATION

### Given Information

A Smart Campus has been allocated the following IP address block:

Network Address: 192.168.100.0/24

The subnet requirements are:

1. School of Computing – /25
2. School of Electronics – /26
3. School of Mechanical Engineering – /27

We need to determine the network address, broadcast address, valid host range, usable host addresses, and remaining unused IP address range.

### Step 1: Understand the Subnetting Concept

IPv4 addresses contain 32 bits. Subnetting is the process of dividing a large network into smaller networks called subnets.

The subnet mask determines the network portion and host portion of an IP address.

The following formulas are used for subnet calculations.

Formula 1: Number of Host Bits

Host Bits = 32 - Prefix Length

Formula 2: Total Number of IP Addresses

Total IP Addresses = 2^(Number of Host Bits)

Formula 3: Number of Usable Host Addresses

Usable Hosts = Total IP Addresses - 2

We subtract 2 because:

- One address is reserved for the network address.
- One address is reserved for the broadcast address.

**Step 2: Calculate the Subnet Sizes**

A. School of Computing – /25

Host Bits = 32 - 25 = 7

Total IP Addresses = 2^7 = 128

Usable Hosts = 128 - 2 = 126

B. School of Electronics – /26

Host Bits = 32 - 26 = 6

Total IP Addresses = 2^6 = 64

Usable Hosts = 64 - 2 = 62

C. School of Mechanical Engineering – /27

Host Bits = 32 - 27 = 5

Total IP Addresses = 2^5 = 32

Usable Hosts = 32 - 2 = 30

**Step 3: Allocate the Subnets**

**A. School of Computing – /25**

Given Network: 192.168.100.0/25

Subnet Mask: 255.255.255.128

Calculations:

Host Bits = 32 - 25 = 7

Total IP Addresses = 2^7 = 128

Usable Hosts = 128 - 2 = 126

Network Address: 192.168.100.0

First Usable Host: 192.168.100.1

Last Usable Host: 192.168.100.126

Broadcast Address: 192.168.100.127

**B. School of Electronics – /26**

The next available address after the Computing subnet is 192.168.100.128.

Given Network: 192.168.100.128/26

Subnet Mask: 255.255.255.192

Calculations:

Host Bits = 32 - 26 = 6

Total IP Addresses = 2^6 = 64

Usable Hosts = 64 - 2 = 62

Network Address: 192.168.100.128

First Usable Host: 192.168.100.129

Last Usable Host: 192.168.100.190

Broadcast Address: 192.168.100.191

**C. School of Mechanical Engineering – /27**

The next available address is 192.168.100.192.

Given Network: 192.168.100.192/27

Subnet Mask: 255.255.255.224

Calculations:

Host Bits = 32 - 27 = 5

Total IP Addresses = 2^5 = 32

Usable Hosts = 32 - 2 = 30

Network Address: 192.168.100.192

First Usable Host: 192.168.100.193

Last Usable Host: 192.168.100.222

Broadcast Address: 192.168.100.223

**Step 4: Final Subnet Allocation Table**

| Department | Network Address | Valid Host Range | Broadcast Address | Usable Hosts |
|---|---|---|---|---|
| Computing | 192.168.100.0 | 192.168.100.1 – 126 | 192.168.100.127 | 126 |
| Electronics | 192.168.100.128 | 192.168.100.129 – 190 | 192.168.100.191 | 62 |
| Mechanical | 192.168.100.192 | 192.168.100.193 – 222 | 192.168.100.223 | 30 |

**Step 5: Calculate the Remaining Unused IP Address Range**

The original network is 192.168.100.0/24.

Host Bits = 32 - 24 = 8

Total IP Addresses = 2^8 = 256

Total Allocated Addresses:

128 + 64 + 32 = 224

Remaining Addresses:

256 - 224 = 32

The allocated subnets end at 192.168.100.223.

Therefore, the remaining unused IP address range is:

Starting Address: 192.168.100.224

Ending Address: 192.168.100.255

Remaining Subnet: 192.168.100.224/27

Total Remaining IP Addresses = 32

Usable Host Addresses = 32 - 2 = 30

**Conclusion**

The Smart Campus network is divided into three subnets according to the department requirements. The School of Computing receives 126 usable host addresses, the School of Electronics receives 62 usable host addresses, and the School of Mechanical Engineering receives 30 usable host addresses. The remaining unused IP address range is 192.168.100.224 to 192.168.100.255.

**SCENARIO 2: MULTINATIONAL SOFTWARE COMPANY – IPv4 SUBNETTING**

**Given Information**

A multinational software company has been allocated the private IP address block:

Network Address: 10.10.0.0/16

The subnet requirements are:

1. Software Development – /24
2. Quality Assurance – /25
3. Human Resources – /26
4. Executive Management – /27

We need to calculate the network address, broadcast address, valid host range, usable host addresses, and remaining unused IP address range.

**Step 1: Calculate the Subnet Sizes**

The formulas used are:

Host Bits = 32 - Prefix Length

Total IP Addresses = 2^(Host Bits)

Usable Hosts = Total IP Addresses - 2

A. Software Development – /24

Host Bits = 32 - 24 = 8

Total IP Addresses = 2^8 = 256

Usable Hosts = 256 - 2 = 254

B. Quality Assurance – /25

Host Bits = 32 - 25 = 7

Total IP Addresses = 2^7 = 128

Usable Hosts = 128 - 2 = 126

C. Human Resources – /26

Host Bits = 32 - 26 = 6

Total IP Addresses = 2^6 = 64

Usable Hosts = 64 - 2 = 62

D. Executive Management – /27

Host Bits = 32 - 27 = 5

Total IP Addresses = 2^5 = 32

Usable Hosts = 32 - 2 = 30

**Step 2: Allocate the Subnets**

**A. Software Development – /24**

Given Network: 10.10.0.0/24

Subnet Mask: 255.255.255.0

Calculations:

Host Bits = 32 - 24 = 8

Total IP Addresses = 2^8 = 256

Usable Hosts = 256 - 2 = 254

Network Address: 10.10.0.0

First Usable Host: 10.10.0.1

Last Usable Host: 10.10.0.254

Broadcast Address: 10.10.0.255

**B. Quality Assurance – /25**

The next available network is 10.10.1.0.

Given Network: 10.10.1.0/25

Subnet Mask: 255.255.255.128

Calculations:

Host Bits = 32 - 25 = 7

Total IP Addresses = 2^7 = 128

Usable Hosts = 128 - 2 = 126

Network Address: 10.10.1.0

First Usable Host: 10.10.1.1

Last Usable Host: 10.10.1.126

Broadcast Address: 10.10.1.127

**C. Human Resources – /26**

The next available network is 10.10.1.128.

Given Network: 10.10.1.128/26

Subnet Mask: 255.255.255.192

Calculations:

Host Bits = 32 - 26 = 6

Total IP Addresses = 2^6 = 64

Usable Hosts = 64 - 2 = 62

Network Address: 10.10.1.128

First Usable Host: 10.10.1.129

Last Usable Host: 10.10.1.190

Broadcast Address: 10.10.1.191

**D. Executive Management – /27**

The next available network is 10.10.1.192.

Given Network: 10.10.1.192/27

Subnet Mask: 255.255.255.224

Calculations:

Host Bits = 32 - 27 = 5

Total IP Addresses = 2^5 = 32

Usable Hosts = 32 - 2 = 30

Network Address: 10.10.1.192

First Usable Host: 10.10.1.193

Last Usable Host: 10.10.1.222

Broadcast Address: 10.10.1.223

**Step 3: Final Subnet Allocation Table**

| Department | Network Address | Valid Host Range | Broadcast Address | Usable Hosts |
|---|---|---|---|---|
| Software Development | 10.10.0.0 | 10.10.0.1 – 254 | 10.10.0.255 | 254 |
| Quality Assurance | 10.10.1.0 | 10.10.1.1 – 126 | 10.10.1.127 | 126 |
| Human Resources | 10.10.1.128 | 10.10.1.129 – 190 | 10.10.1.191 | 62 |
| Executive Management | 10.10.1.192 | 10.10.1.193 – 222 | 10.10.1.223 | 30 |

**Step 4: Calculate the Remaining Unused IP Address Range**

The original network is 10.10.0.0/16.

Host Bits = 32 - 16 = 16

Total IP Addresses = 2^16 = 65,536

Total Allocated Addresses:

256 + 128 + 64 + 32 = 480

Remaining Addresses:

65,536 - 480 = 65,056

The last allocated address is 10.10.1.223.

Therefore, the remaining unused IP address range is:

Starting Address: 10.10.1.224

Ending Address: 10.10.255.255

Total Remaining IP Addresses = 65,056

This unused range can be divided into multiple CIDR blocks for future department expansion.

**Conclusion**

The company network is divided into four subnets based on department requirements. Software Development receives 254 usable host addresses, Quality Assurance receives 126, Human Resources receives 62, and Executive Management receives 30. A total of 65,056 IP addresses remain unused within the allocated /16 network block.

**SCENARIO 3: SMART HOSPITAL – IPv6 ADDRESSING**

**Given Information**

A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors.

The allocated IPv6 network is:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

We need to perform IPv6 compression, expansion, identify the network prefix, and calculate the total number of possible host address combinations.

**Step 1: Understand IPv6 Addressing**

IPv6 is the latest version of the Internet Protocol. It uses 128-bit addresses instead of the 32-bit addresses used in IPv4.

An IPv6 address consists of:

1. 128 bits.
2. Eight groups called hextets.
3. Each hextet contains 16 bits.
4. Each hextet is represented using four hexadecimal digits.

Example:

2001:0db8:abcd:0000:0000:0000:0000:0000

**Step 2: Convert the IPv6 Address into Compressed Notation**

IPv6 compression follows two important rules.

Rule 1: Remove Leading Zeros

Leading zeros in each hextet can be removed.

Examples:

0db8 becomes db8.

0000 becomes 0.

After removing leading zeros:

2001:db8:abcd:0:0:0:0:0

Rule 2: Replace Consecutive Zero Groups with Double Colon (::)

A continuous sequence of zero hextets can be replaced with a double colon (::).

Therefore, the compressed IPv6 address is:

2001:db8:abcd::/64

**Step 3: Expand the Compressed IPv6 Address**

Given compressed address:

2001:db8:abcd::/64

The IPv6 address must contain eight hextets.

The double colon represents five missing zero hextets.

Therefore, the expanded IPv6 address is:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

**Step 4: Identify the Network Prefix Using /64**

Given prefix length: /64

An IPv6 address contains 128 bits.

The /64 prefix means:

- First 64 bits represent the network portion.
- Remaining 64 bits represent the interface identifier.

The first four hextets represent the network prefix.

Network Prefix:

2001:0db8:abcd:0000/64

Network Bits = 64 bits

Interface Identifier Bits = 64 bits

**Step 5: Calculate the Total Number of Possible Host Addresses**

Formula:

Host Bits = 128 - Prefix Length

Substitution:

Host Bits = 128 - 64

Host Bits = 64

Total Possible Interface Identifier Combinations = 2^64

2^64 = 18,446,744,073,709,551,616

Therefore, the /64 IPv6 subnet supports:

18,446,744,073,709,551,616 possible interface identifier combinations.

Note: IPv6 does not use broadcast addresses. The above value represents the mathematical number of possible interface identifier combinations, not a guarantee that every combination is an ordinary assignable unicast host address.

**Step 6: Final IPv6 Answers**

1. Original IPv6 Address:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

2. Compressed Notation:

2001:db8:abcd::/64

3. Expanded Notation:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

4. Network Prefix:

2001:0db8:abcd:0000/64

5. Network Bits:

64 bits

6. Interface Identifier Bits:

64 bits

7. Total Possible Interface Identifier Combinations:

2^64 = 18,446,744,073,709,551,616

**Conclusion**

The given IPv6 address can be compressed into 2001:db8:abcd::/64. Its first 64 bits represent the network prefix, and the remaining 64 bits represent the

interface identifier. Therefore, the network supports 2^64 possible interface identifier combinations, providing a very large address space for the hospital's connected medical devices and IoT sensors.

## SCENARIO 4: ISP CORE ROUTER – ROUTING TABLE AND PACKET FORWARDING

### Given Information

An Internet Service Provider connects multiple customer networks through a core router.

The router maintains the following routing table entries:

1. 192.168.1.0/24
2. 192.168.2.0/24
3. 192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

We need to determine the destination subnet, matching routing table entry, next-hop interface, and default route.

### Step 1: Understand Routing

Routing is the process of selecting the best path for forwarding packets from a source network to a destination network.

A router examines the destination IP address of an incoming packet and compares it with the entries stored in its routing table.

The router uses a technique called Longest Prefix Match (LPM) to select the most specific matching route.

### Step 2: Determine the Destination Subnet

Given Destination IP Address:

192.168.2.45

Routing Table:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A /24 subnet has the subnet mask:

255.255.255.0

The first three octets identify the network portion.

Compare the destination address with the routing table:

1. 192.168.1.0/24 – No match because the third octet is 1.
2. 192.168.2.0/24 – Match because the first three octets are the same.
3. 192.168.3.0/24 – No match because the third octet is 3.

Therefore, the destination subnet is:

192.168.2.0/24

**Step 3: Identify the Matching Routing Table Entry**

The matching routing table entry is:

192.168.2.0/24

The destination address 192.168.2.45 belongs to this subnet.

The subnet details are:

Network Address: 192.168.2.0

First Usable Host: 192.168.2.1

Last Usable Host: 192.168.2.254

Broadcast Address: 192.168.2.255

Since 192.168.2.45 lies within the valid host range, it belongs to the 192.168.2.0/24 network.

**Step 4: Determine the Next-Hop Network/Interface**

The router forwards the packet through the interface connected to the 192.168.2.0/24 network.

For example, assume the router has the following interfaces:

GigabitEthernet0/0 – Connected to 192.168.1.0/24

GigabitEthernet0/1 – Connected to 192.168.2.0/24

GigabitEthernet0/2 – Connected to 192.168.3.0/24

In this example, the packet is forwarded through:

GigabitEthernet0/1

Important: The question does not provide actual router interface names or next-hop IP addresses. Therefore, the exact interface cannot be determined from the routing table alone. GigabitEthernet0/1 is only an example.

If the destination network is directly connected, the router forwards the packet through the connected interface after resolving the destination MAC address using ARP.

**Step 5: Determine the Default Route**

A default route is used when the destination IP address does not match any specific routing table entry.

The IPv4 default route is:

0.0.0.0/0

It is also known as the Gateway of Last Resort.

Example Cisco Router Command:

Router(config)# ip route 0.0.0.0 0.0.0.0 10.10.10.1

Explanation:

- 0.0.0.0 represents any destination network.
- 0.0.0.0 represents the default subnet mask.
- 10.10.10.1 is an example of the next-hop router IP address.

**Step 6: Final Answers**

1. Destination IP Address:

192.168.2.45

2. Destination Subnet:

192.168.2.0/24

3. Matching Routing Table Entry:

192.168.2.0/24

4. Next-Hop Network/Interface:

The interface connected to 192.168.2.0/24.

Example: GigabitEthernet0/1.

5. Default Route:

0.0.0.0/0

**Conclusion**

The router identifies that the destination IP address 192.168.2.45 belongs to the 192.168.2.0/24 subnet. Therefore, it selects the matching routing table entry and forwards the packet through the interface connected to that network. If the destination does not match any specific routing entry, the router uses the default route 0.0.0.0/0, provided it is configured.

**SCENARIO 5: TWO OFFICE DEPARTMENTS – LAN SUBNETTING AND DEFAULT GATEWAY CONFIGURATION**

**Given Information**

A company establishes two office departments connected through a Layer-3 router.

The allocated subnet information is:

Administration LAN – 192.168.10.0/26

Finance LAN – 192.168.10.64/26

Router Interface Configuration:

Administration Gateway – 192.168.10.1

Finance Gateway – 192.168.10.65

We need to determine the network address, broadcast address, valid host range, suitable IP addresses for three computers in each LAN, and the default gateway.

**Step 1: Understand the Subnet Mask**

Given Prefix Length: /26

IPv4 Address Size = 32 bits

Formula 1: Number of Host Bits

Host Bits = 32 - Prefix Length

Host Bits = 32 - 26 = 6

Formula 2: Total Number of IP Addresses

Total IP Addresses = 2^6 = 64

Formula 3: Usable Host Addresses

Usable Hosts = 64 - 2 = 62

Therefore, each /26 subnet contains:

Total IP Addresses = 64

Usable Host Addresses = 62

Subnet Mask = 255.255.255.192

**Step 2: Calculate the Block Size**

Formula:

Block Size = 256 - Subnet Mask Value in the Interesting Octet

The interesting octet is the fourth octet.

Block Size = 256 - 192

Block Size = 64

Therefore, the subnet network addresses increase in intervals of 64.

The network addresses are:

192.168.10.0

192.168.10.64

192.168.10.128

192.168.10.192

The given Administration and Finance LAN addresses are valid /26 network addresses.

**Step 3: Administration LAN Calculation**

Given Network:

192.168.10.0/26

Subnet Mask:

255.255.255.192

Total IP Addresses:

2^(32 - 26) = 2^6 = 64

Usable Host Addresses:

64 - 2 = 62

Network Address:

192.168.10.0

Broadcast Address Calculation:

0 + 64 - 1 = 63

Broadcast Address:

192.168.10.63

Valid Host Range:

First Usable Host: 192.168.10.1

Last Usable Host: 192.168.10.62

Default Gateway:

192.168.10.1

**Step 4: Finance LAN Calculation**

Given Network:

192.168.10.64/26

Subnet Mask:

255.255.255.192

Total IP Addresses:

2^(32 - 26) = 2^6 = 64

Usable Host Addresses:

64 - 2 = 62

Network Address:

192.168.10.64

Broadcast Address Calculation:

64 + 64 - 1 = 127

Broadcast Address:

192.168.10.127

Valid Host Range:

First Usable Host: 192.168.10.65

Last Usable Host: 192.168.10.126

Default Gateway:

192.168.10.65

**Step 5: Suggest Suitable IP Addresses for Three Computers in Each LAN**

A. Administration LAN

The Administration LAN uses the network 192.168.10.0/26.

Default Gateway: 192.168.10.1

Suitable computer IP addresses:

Computer 1 – 192.168.10.2

Computer 2 – 192.168.10.3

Computer 3 – 192.168.10.4

Subnet Mask for All Computers:

255.255.255.192

Default Gateway for All Computers:

192.168.10.1

B. Finance LAN

The Finance LAN uses the network 192.168.10.64/26.

Default Gateway: 192.168.10.65

Suitable computer IP addresses:

Computer 1 – 192.168.10.66

Computer 2 – 192.168.10.67

Computer 3 – 192.168.10.68

Subnet Mask for All Computers:

255.255.255.192

Default Gateway for All Computers:

192.168.10.65

**Step 6: Final Computer IP Allocation Table**

| LAN / Device | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|
| Administration PC1 | 192.168.10.2 | 255.255.255.192 | 192.168.10.1 |
| Administration PC2 | 192.168.10.3 | 255.255.255.192 | 192.168.10.1 |
| Administration PC3 | 192.168.10.4 | 255.255.255.192 | 192.168.10.1 |
| Finance PC1 | 192.168.10.66 | 255.255.255.192 | 192.168.10.65 |
| Finance PC2 | 192.168.10.67 | 255.255.255.192 | 192.168.10.65 |
| Finance PC3 | 192.168.10.68 | 255.255.255.192 | 192.168.10.65 |

**Step 7: Identify the Role of the Default Gateway**

A default gateway is the router interface address that a computer uses to communicate with devices outside its local network.

Administration LAN:

All Administration computers use 192.168.10.1 as their default gateway.

Finance LAN:

All Finance computers use 192.168.10.65 as their default gateway.

For example, when an Administration computer wants to communicate with a Finance computer, it sends the packet to its default gateway. The Layer-3 router examines the destination IP address and forwards the packet to the Finance LAN.

This process enables communication between the two different networks.

**Step 8: Final Subnet Allocation Table**

| Parameter | Administration LAN | Finance LAN |
|---|---|---|
| Network Address | 192.168.10.0 | 192.168.10.64 |
| Prefix Length | /26 | /26 |
| Subnet Mask | 255.255.255.192 | 255.255.255.192 |
| Broadcast Address | 192.168.10.63 | 192.168.10.127 |
| First Usable Host | 192.168.10.1 | 192.168.10.65 |
| Last Usable Host | 192.168.10.62 | 192.168.10.126 |
| Total IP Addresses | 64 | 64 |
| Usable Host Addresses | 62 | 62 |

| | | |
|---|---|---|
| Default Gateway | 192.168.10.1 | 192.168.10.65 |

**Conclusion**

Both Administration and Finance LANs use /26 subnet masks. Each LAN contains 64 total IP addresses and supports 62 usable host addresses. The Administration computers use 192.168.10.1 as their default gateway, while the Finance computers use 192.168.10.65. The Layer-3 router enables communication between the two departments.